

Exploratory Data Analysis

Understanding the dataset to explore how the data is present in the database and if there is a need of creating some aggregated tables that can help with:

- Vendor selection for profitability
- Product Pricing Optimization

```
import pandas as pd
import sqlite3

# create database connection
conn = sqlite3.connect('inventory.db')

# checking tables present in the database
tables = pd.read_sql_query('select name from sqlite_master where
                             type="table"', conn)

tables
```

	name
0	begin_inventory
1	end_inventory
2	purchases
3	purchase_prices
4	sales
5	vendor_invoice

```
for table in tables['name']:
    print('-'*50, f'{table}', '-'*50)
    print('Count of records:', pd.read_sql('select count(*) as count from
        purchases', conn)['count'].values[0])
    display(pd.read_sql(f'select * from {table} limit 5', conn))
```

```
----- begin_inventory -----
-----
Count of records: 2372474
```

	InventoryId	Store	City	Brand	Description	
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	75
1	1_HARDERSFIELD_60	1	HARDERSFIELD	60	Canadian Club 1858 VAP	75
2	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	75

	InventoryId	Store	City	Brand	Description	
3	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	75
4	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	75

----- end_inventory -----

Count of records: 2372474

	InventoryId	Store	City	Brand	Description	
0	1_HARDERSFIELD_58	1	HARDERSFIELD	58	Gekkeikan Black & Gold Sake	75
1	1_HARDERSFIELD_62	1	HARDERSFIELD	62	Herradura Silver Tequila	75
2	1_HARDERSFIELD_63	1	HARDERSFIELD	63	Herradura Reposado Tequila	75
3	1_HARDERSFIELD_72	1	HARDERSFIELD	72	No. 3 London Dry Gin	75
4	1_HARDERSFIELD_75	1	HARDERSFIELD	75	Three Olives Tomato Vodka	75

----- purchases -----

Count of records: 2372474

	InventoryId	Store	Brand	Description	Size	VendorNu
0	69_MOUNTMEND_8412	69	8412	Tequila Ocho Plata Fresno	750mL	105
1	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466
2	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466
3	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466
4	76_DONCASTER_2034	76	2034	Glendalough Double Barrel	750mL	388

----- purchase_prices -----

Count of records: 2372474

	Brand	Description	Price	Size	Volume	Classification	PurchasePrice
0	58	Gekkeikan Black & Gold Sake	12.99	750mL	750	1	9.28
1	62	Herradura Silver Tequila	36.99	750mL	750	1	28.67
2	63	Herradura Reposado Tequila	38.99	750mL	750	1	30.46
3	72	No. 3 London Dry Gin	34.99	750mL	750	1	26.11
4	75	Three Olives Tomato Vodka	14.99	750mL	750	1	10.94

----- sales -----

Count of records: 2372474

	InventoryId	Store	Brand	Description	Size	SalesQuantity
0	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1
1	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	2
2	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1
3	1_HARDERSFIELD_1004	1	1004	Jim Beam w/2 Rocks Glasses	750mL	1
4	1_HARDERSFIELD_1005	1	1005	Maker's Mark Combo Pack	375mL 2 Pk	2

----- vendor_invoice -----

Count of records: 2372474

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDate
0	105	ALTAMAR BRANDS LLC	2024-01-04	8124	2023- 12-21	2024- 02-16
1	4466	AMERICAN VINTAGE	2024-01-07	8137	2023- 12-22	2024- 02-21

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayDa
		BEVERAGE				
2	388	ATLANTIC IMPORTING COMPANY	2024-01-09	8169	2023- 12-24	2024- 02-16
3	480	BACARDI USA INC	2024-01-12	8106	2023- 12-20	2024- 02-05
4	516	BANFI PRODUCTS CORP	2024-01-07	8170	2023- 12-24	2024- 02-12

```
purchases = pd.read_sql_query('select * from purchases where VendorNumber =
4466', conn)
```

purchases

	InventoryId	Store	Brand	Description	Size	VendorN
0	30_CULCHETH_5255	30	5255	TGI Fridays Ultimte Mudslide	1.75L	4466
1	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466
2	1_HARDERSFIELD_5255	1	5255	TGI Fridays Ultimte Mudslide	1.75L	4466
3	38_GOULCREST_5215	38	5215	TGI Fridays Long Island Iced	1.75L	4466
4	59_CLAETHORPES_5215	59	5215	TGI Fridays Long Island Iced	1.75L	4466
...	...	...	...	...	...	...
2187	81_PEMBROKE_5215	81	5215	TGI Fridays Long Island Iced	1.75L	4466
2188	62_KILMARNOCK_5255	62	5255	TGI Fridays Ultimte Mudslide	1.75L	4466
2189	34_PITMERDEN_5215	34	5215	TGI Fridays Long Island Iced	1.75L	4466
2190	6_GOULCREST_5215	6	5215	TGI Fridays Long Island Iced	1.75L	4466
2191	35_HALIVAARA_5255	35	5255	TGI Fridays Ultimte Mudslide	1.75L	4466

2192 rows × 16 columns

```
purchase_price = pd.read_sql_query('''select * from purchase_prices where
VendorNumber = 4466''', conn)
```

purchase_price

	Brand	Description	Price	Size	Volume	Classification	PurchasePri
0	5215	TGI Fridays Long Island Iced	12.99	1750mL	1750	1	9.41
1	5255	TGI Fridays Ultimte Mudslide	12.99	1750mL	1750	1	9.35
2	3140	TGI Fridays Orange Dream	14.99	1750mL	1750	1	11.19

```
vendor_invoice = pd.read_sql_query('''select * from vendor_invoice where
VendorNumber = 4466''', conn)
```

vendor_invoice

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayD
0	4466	AMERICAN VINTAGE BEVERAGE	2024-01-07	8137	2023-12-22	2024-02-21
1	4466	AMERICAN VINTAGE BEVERAGE	2024-01-19	8207	2023-12-27	2024-02-26
2	4466	AMERICAN VINTAGE BEVERAGE	2024-01-18	8307	2024-01-03	2024-02-18
3	4466	AMERICAN VINTAGE BEVERAGE	2024-01-27	8469	2024-01-14	2024-03-11
4	4466	AMERICAN VINTAGE BEVERAGE	2024-02-04	8532	2024-01-19	2024-03-15
5	4466	AMERICAN VINTAGE BEVERAGE	2024-02-09	8604	2024-01-24	2024-03-15
6	4466	AMERICAN VINTAGE BEVERAGE	2024-02-17	8793	2024-02-05	2024-04-02
7	4466	AMERICAN VINTAGE BEVERAGE	2024-03-01	8892	2024-02-12	2024-03-28
8	4466	AMERICAN VINTAGE BEVERAGE	2024-03-07	8995	2024-02-19	2024-04-02
9	4466	AMERICAN VINTAGE BEVERAGE	2024-03-12	9033	2024-02-22	2024-04-16
10	4466	AMERICAN VINTAGE BEVERAGE	2024-03-16	9180	2024-03-03	2024-04-19

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayD
11	4466	AMERICAN VINTAGE BEVERAGE	2024-03-23	9244	2024-03-08	2024-04-21
12	4466	AMERICAN VINTAGE BEVERAGE	2024-03-31	9371	2024-03-17	2024-05-13
13	4466	AMERICAN VINTAGE BEVERAGE	2024-04-09	9491	2024-03-24	2024-05-08
14	4466	AMERICAN VINTAGE BEVERAGE	2024-04-17	9583	2024-03-31	2024-05-12
15	4466	AMERICAN VINTAGE BEVERAGE	2024-04-20	9639	2024-04-04	2024-06-04
16	4466	AMERICAN VINTAGE BEVERAGE	2024-04-29	9800	2024-04-15	2024-06-07
17	4466	AMERICAN VINTAGE BEVERAGE	2024-05-09	9886	2024-04-21	2024-06-12
18	4466	AMERICAN VINTAGE BEVERAGE	2024-05-14	9999	2024-04-29	2024-06-26
19	4466	AMERICAN VINTAGE BEVERAGE	2024-05-16	10095	2024-05-06	2024-06-27
20	4466	AMERICAN VINTAGE BEVERAGE	2024-05-28	10169	2024-05-11	2024-07-04
21	4466	AMERICAN VINTAGE BEVERAGE	2024-06-04	10257	2024-05-17	2024-07-08
22	4466	AMERICAN VINTAGE BEVERAGE	2024-06-12	10346	2024-05-23	2024-07-20
23	4466	AMERICAN VINTAGE BEVERAGE	2024-06-17	10445	2024-05-30	2024-07-19
24	4466	AMERICAN VINTAGE BEVERAGE	2024-06-22	10600	2024-06-09	2024-08-01
25	4466	AMERICAN VINTAGE BEVERAGE	2024-06-29	10695	2024-06-16	2024-08-12
26	4466	AMERICAN VINTAGE BEVERAGE	2024-07-09	10777	2024-06-22	2024-08-15
27	4466	AMERICAN VINTAGE BEVERAGE	2024-07-08	10836	2024-06-25	2024-08-14

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayD
28	4466	AMERICAN VINTAGE BEVERAGE	2024-07-11	10969	2024-06-29	2024-08-18
29	4466	AMERICAN VINTAGE BEVERAGE	2024-07-19	11085	2024-07-06	2024-09-04
30	4466	AMERICAN VINTAGE BEVERAGE	2024-07-26	11187	2024-07-14	2024-08-26
31	4466	AMERICAN VINTAGE BEVERAGE	2024-08-03	11244	2024-07-18	2024-09-04
32	4466	AMERICAN VINTAGE BEVERAGE	2024-08-11	11362	2024-07-26	2024-09-24
33	4466	AMERICAN VINTAGE BEVERAGE	2024-08-18	11489	2024-08-04	2024-09-16
34	4466	AMERICAN VINTAGE BEVERAGE	2024-08-24	11540	2024-08-08	2024-10-02
35	4466	AMERICAN VINTAGE BEVERAGE	2024-09-02	11716	2024-08-19	2024-09-29
36	4466	AMERICAN VINTAGE BEVERAGE	2024-09-12	11771	2024-08-23	2024-10-11
37	4466	AMERICAN VINTAGE BEVERAGE	2024-09-20	11901	2024-09-01	2024-10-30
38	4466	AMERICAN VINTAGE BEVERAGE	2024-09-25	11993	2024-09-07	2024-10-23
39	4466	AMERICAN VINTAGE BEVERAGE	2024-10-01	12125	2024-09-16	2024-11-07
40	4466	AMERICAN VINTAGE BEVERAGE	2024-10-08	12235	2024-09-23	2024-11-20
41	4466	AMERICAN VINTAGE BEVERAGE	2024-10-09	12253	2024-09-23	2024-11-14
42	4466	AMERICAN VINTAGE BEVERAGE	2024-10-12	12321	2024-09-26	2024-11-19
43	4466	AMERICAN VINTAGE BEVERAGE	2024-10-20	12466	2024-10-05	2024-11-26
44	4466	AMERICAN VINTAGE BEVERAGE	2024-10-27	12515	2024-10-09	2024-11-30

	VendorNumber	VendorName	InvoiceDate	PONumber	PODate	PayD
45	4466	AMERICAN VINTAGE BEVERAGE	2024-11-07	12702	2024-10-21	2024-12-11
46	4466	AMERICAN VINTAGE BEVERAGE	2024-11-12	12752	2024-10-25	2024-12-11
47	4466	AMERICAN VINTAGE BEVERAGE	2024-11-20	12828	2024-10-30	2024-12-18
48	4466	AMERICAN VINTAGE BEVERAGE	2024-11-27	12929	2024-11-06	2025-01-04
49	4466	AMERICAN VINTAGE BEVERAGE	2024-11-28	13092	2024-11-16	2024-12-30
50	4466	AMERICAN VINTAGE BEVERAGE	2024-12-06	13134	2024-11-20	2025-01-18
51	4466	AMERICAN VINTAGE BEVERAGE	2024-12-16	13254	2024-11-28	2025-01-13
52	4466	AMERICAN VINTAGE BEVERAGE	2024-12-26	13432	2024-12-09	2025-01-27
53	4466	AMERICAN VINTAGE BEVERAGE	2024-12-30	13483	2024-12-13	2025-02-11
54	4466	AMERICAN VINTAGE BEVERAGE	2025-01-09	13627	2024-12-22	2025-02-05

```
sales = pd.read_sql_query(''select * from sales where VendorNo = 4466'',
                           conn)
sales
```

	InventoryId	Store	Brand	Description	Size	SalesQu
0	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1
1	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1
2	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1
3	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1
4	1_HARDERSFIELD_5215	1	5215	TGI Fridays Long Island Iced	1.75L	1

	InventoryId	Store	Brand	Description	Size	SalesQua
...	...	...	...	...	...	...
9448	9_BLACKPOOL_5215	9	5215	TGI Fridays Long Island Iced	1.75L	1
9449	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1
9450	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1
9451	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1
9452	9_BLACKPOOL_5255	9	5255	TGI Fridays Ultimte Mudslide	1.75L	1

9453 rows × 14 columns

```
purchases.groupby(['Brand', 'PurchasePrice'])[['Quantity', 'Dollars']].sum()
```

		Quantity	Dollars
Brand	PurchasePrice		
3140	11.19	4640	51921.60
5215	9.41	4923	46325.43
5255	9.35	6215	58110.25

purchase_price

	Brand	Description	Price	Size	Volume	Classification	PurchasePr
0	5215	TGI Fridays Long Island Iced	12.99	1750mL	1750	1	9.41
1	5255	TGI Fridays Ultimte Mudslide	12.99	1750mL	1750	1	9.35
2	3140	TGI Fridays Orange Dream	14.99	1750mL	1750	1	11.19

```
sales.groupby('Brand')[['SalesDollars', 'SalesPrice', 'SalesQuantity']].sum()
```

	SalesDollars	SalesPrice	SalesQuantity
Brand			
3140	50531.10	30071.85	3890
5215	60416.49	41542.02	4651
5255	79187.04	51180.60	6096

- The purchases table stores detailed transaction data, including the purchase date, products (brands) bought by vendors, the amount paid (in dollars), and the quantity purchased.
- The purchase_prices table provides product-level pricing details, including both actual and purchase prices. In this table, each combination of vendor and brand is unique.
- The vendor_invoice table summarizes data from the purchases table by aggregating quantities and total amounts, and also includes freight costs. Each record is uniquely identified by the vendor and PO number.
- The sales table records sales transactions, capturing the brands sold by vendors, quantities sold, selling prices, and the revenue generated.

Since the data required for analysis is spread across multiple tables, we need to create a consolidated summary table that includes:

- Purchase transactions made by vendors
- Sales transaction details
- Freight costs associated with each vendor
- Actual product prices from vendors

vendor_invoice.columns

```
Index(['VendorNumber', 'VendorName', 'InvoiceDate', 'PONumber', 'PODate',
      'PayDate', 'Quantity', 'Dollars', 'Freight', 'Approval'],
      dtype='object')
```

```
freight_summary = pd.read_sql_query("""select VendorNumber, sum(Freight) as
      Freight_Cost
from vendor_invoice
group by VendorNumber""", conn)
freight_summary
```

	VendorNumber	Freight_Cost
0	2	27.08
1	54	0.48
2	60	367.52
3	105	62.39
4	200	6.19
...	...	...
121	98450	856.02
122	99166	130.09
123	172662	178.34
124	173357	202.50
125	201359	0.09

126 rows × 2 columns

```
pd.read_sql_query("""select
      p.VendorNumber,
      p.VendorName,
      p.Brand,
      p.PurchasePrice,
      pp.Volume,
```

```

pp.price as ActualPrices,
sum(quantity) as TotalPurchaseQuantity,
sum(Dollars) as TotalPurchaseDollars
from purchases p
JOIN purchase_prices pp
on p.Brand = pp.Brand
where p.PurchasePrice > 0
group by p.VendorNumber, p.VendorName, p.Brand
Order by TotalPurchaseDollars""", conn)

```

	VendorNumber	VendorName	Brand	PurchasePrice	Volume	Ac
0	7245	PROXIMO SPIRITS INC.	3065	0.71	50	0.9
1	3960	DIAGEO NORTH AMERICA INC	6127	1.47	200	1.9
2	3924	HEAVEN HILL DISTILLERIES	9123	0.74	50	0.9
3	8004	SAZERAC CO INC	5683	0.39	50	0.4
4	9815	WINE GROUP INC	8527	1.32	750	4.9
...	...	...	...	...	...	...
10687	3960	DIAGEO NORTH AMERICA INC	3545	21.89	1750	29
10688	3960	DIAGEO NORTH AMERICA INC	4261	16.17	1750	22
10689	17035	PERNOD RICARD USA	8068	18.24	1750	24
10690	4425	MARTIGNETTI COMPANIES	3405	23.19	1750	28
10691	1128	BROWN-FORMAN CORP	1233	26.27	1750	36

10692 rows × 8 columns

sales.columns

```

Index(['InventoryId', 'Store', 'Brand', 'Description', 'Size',
      'SalesQuantity',
      'SalesDollars', 'SalesPrice', 'SalesDate', 'Volume', 'Classification',
      'ExciseTax', 'VendorNo', 'VendorName'],
      dtype='object')

```

```

pd.read_sql_query("""select
    VendorNo,
    Brand,
    SUM(SalesDollars) as TotalSalesDollars,
    SUM(SalesPrice) as TotalSalesPrice,
    SUM(SalesQuantity) as TotalSalesQuantity,
    SUM(ExciseTax) as TotalExciseTax

```

```

from sales
group by VendorNo, Brand
order by TotalSalesDollars""", conn)

```

	VendorNo	Brand	TotalSalesDollars	TotalSalesPrice	TotalSalesQuar
0	8004	5287	0.98	0.98	2
1	9206	2773	0.99	0.99	1
2	3252	3933	1.98	0.99	2
3	3924	9123	1.98	0.99	2
4	10050	3623	1.98	1.98	2
...	...	...	...	...	...
11267	3960	3545	4223107.62	545778.28	135838
11268	3960	4261	4475972.88	420050.01	200412
11269	17035	8068	4538120.60	461140.15	187140
11270	4425	3405	4819073.49	561512.37	160247
11271	1128	1233	5101919.51	672819.31	142049

11272 rows × 6 columns

merging all the summary table to create an aggregated summary table for further analysis

```

vendor_sales_summary = pd.read_sql_query("""WITH FreightSummary AS (
    SELECT
        VendorNumber,
        SUM(Freight) AS FreightCost
    FROM vendor_invoice
    GROUP BY VendorNumber
),

```

```

PurchaseSummary AS (
    SELECT
        p.VendorNumber,
        p.VendorName,
        p.Brand,
        p.Description,
        p.PurchasePrice,
        pp.Price AS ActualPrice,
        pp.Volume,
        SUM(p.Quantity) AS TotalPurchaseQuantity,
        SUM(p.Dollars) AS TotalPurchaseDollars
    FROM purchases p
    JOIN purchase_prices pp
        ON p.Brand = pp.Brand
    WHERE p.PurchasePrice > 0
    GROUP BY p.VendorNumber, p.VendorName, p.Brand, p.Description,
        p.PurchasePrice, pp.Price, pp.Volume
),

```

```

SalesSummary AS (
    SELECT
        VendorNo,
        Brand,

```

```

SUM(SalesQuantity) AS TotalSalesQuantity,
SUM(SalesDollars) AS TotalSalesDollars,
SUM(SalesPrice) AS TotalSalesPrice,
SUM(ExciseTax) AS TotalExciseTax
FROM sales
GROUP BY VendorNo, Brand
)

```

```

SELECT
    ps.VendorNumber,
    ps.VendorName,
    ps.Brand,
    ps.Description,
    ps.PurchasePrice,
    ps.ActualPrice,
    ps.Volume,
    ps.TotalPurchaseQuantity,
    ps.TotalPurchaseDollars,
    ss.TotalSalesQuantity,
    ss.TotalSalesDollars,
    ss.TotalSalesPrice,
    ss.TotalExciseTax,
    fs.FreightCost
FROM PurchaseSummary ps
LEFT JOIN SalesSummary ss
    ON ps.VendorNumber = ss.VendorNo
    AND ps.Brand = ss.Brand
LEFT JOIN FreightSummary fs
    ON ps.VendorNumber = fs.VendorNumber
ORDER BY ps.TotalPurchaseDollars DESC""",conn)

```

vendor_sales_summary

	VendorNumber	VendorName	Brand	Description	PurchasePri
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89
...	...	...	...	...	...
10687	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's	0.39

	VendorNumber	VendorName	Brand	Description	PurchasePri
				Apple Pie	
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71

10692 rows × 14 columns

Performance Optimization:

- It performs complex joins and aggregations on large datasets such as sales and purchases.
- Pre-aggregating and storing the results reduces the need for repeated, resource-intensive computations.
- Enables efficient analysis of sales performance, purchase trends, and pricing across vendors and brands.
- Improves future dashboarding and reporting by providing readily available summarized data.
- Dashboards can retrieve data directly from vendor_sales_summary instead of executing heavy queries repeatedly.

vendor_sales_summary.dtypes

```
VendorNumber      int64
VendorName        object
Brand             int64
Description        object
PurchasePrice     float64
ActualPrice       float64
Volume           object
TotalPurchaseQuantity  int64
TotalPurchaseDollars  float64
TotalSalesQuantity  float64
TotalSalesDollars    float64
TotalSalesPrice     float64
TotalExciseTax      float64
FreightCost        float64
dtype: object
```

vendor_sales_summary.isnull().sum()

```
VendorNumber      0
VendorName        0
Brand             0
Description        0
PurchasePrice     0
ActualPrice       0
Volume           0
TotalPurchaseQuantity  0
TotalPurchaseDollars  0
TotalSalesQuantity 178
TotalSalesDollars   178
TotalSalesPrice    178
TotalExciseTax     178
```

```
FreightCost          0
dtype: int64
```

```
vendor_sales_summary.head()
```

	VendorNumber	VendorName	Brand	Description	PurchasePrice	Ac
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.

```
vendor_sales_summary['VendorName'].unique()
```

```
array(['BROWN-FORMAN CORP', 'MARTIGNETTI COMPANIES', 'PERNOD RICARD USA',
      'DIAGEO NORTH AMERICA INC', 'BACARDI USA INC',
      'JIM BEAM BRANDS COMPANY', 'MAJESTIC FINE WINES',
      'ULTRA BEVERAGE COMPANY LLP', 'STOLI GROUP,(USA) LLC',
      'PROXIMO SPIRITS INC.', 'MOET HENNESSY USA INC', 'CAMPARI AMERICA',
      'SAZERAC CO INC', 'CONSTELLATION BRANDS INC', 'M S WALKER INC',
      'SAZERAC NORTH AMERICA INC.', 'PALM BAY INTERNATIONAL INC',
      'REMY COINTREAU USA INC', 'SIDNEY FRANK IMPORTING CO',
      'E & J GALLO WINERY', 'WILLIAM GRANT & SONS INC',
      'HEAVEN HILL DISTILLERIES', 'DISARONNO INTERNATIONAL LLC',
      'EDRINGTON AMERICAS', 'CASTLE BRANDS CORP.',
      'SOUTHERN WINE & SPIRITS NE', 'STE MICHELLE WINE ESTATES',
      'TRINCHERO FAMILY ESTATES', 'MHW LTD', 'WINE GROUP INC',
      'PERFECTA WINES', 'LUXCO INC', 'TREASURY WINE ESTATES',
      'DIAGEO CHATEAU ESTATE WINES', 'SHAW ROSS INT L IMP LTD',
      'PINE STATE TRADING CO', 'PHILLIPS PRODUCTS CO.',
      'CALEDONIA SPIRITS INC', 'STATE WINE & SPIRITS',
      'KOBRA BRAND CORPORATION', 'BANFI PRODUCTS CORP',
      'VINEYARD BRANDS INC', 'DELICATO VINEYARDS INC',
      'FABRIZIA SPIRITS LLC', 'DUGGANS DISTILLED PRODUCTS',
      'Serralles Usa LLC', 'SEA HAGG DISTILLERY LLC',
      'OLE SMOKY DISTILLERY LLC', 'VRANKEN AMERICA', 'KLIN SPIRITS LLC',
      'LAIRD & CO', 'ADAMBA IMPORTS INTL INC',
      'LATITUDE BEVERAGE COMPANY', 'FREDERICK WILDMAN & SONS',
      'MCCORMICK DISTILLING CO', 'CHARLES JACQUIN ET CIE INC',
      'WESTERN SPIRITS BEVERAGE CO', 'MARSALLE COMPANY',
      'AMERICAN VINTAGE BEVERAGE', 'MANGO BOTTLING INC',
      'SWEET BABY VINEYARD', 'NICHE W & S', 'LABELLE VYDS AND WINERY',
      'FLAG HILL WINERY & VINEYARD', 'SMOKY QUARTZ DISTILLERY LLC',
      'PREMIUM PORT WINES, INC.', 'Russian Standard Vodka',
      'Dunn Wine Brokers', 'WEIN BAUER INC', 'BULLY BOY DISTILLERS',
      'ATLANTIC IMPORTING COMPANY', 'PREMIER DISTRIBUTORS',
      'VINILANDIA USA', 'PARK STREET IMPORTS LLC', 'TAKARA SAKE USA INC',
      'SEA BREEZE CELLARS LLC', 'STARK BREWING COMPANY', 'TY KU LLC',
```

```

'PSP WINES', 'TAMWORTH DISTILLING', 'ZORVINO VINEYARDS',
'SOUTHERN GLAZERS W&S OF NE', 'HOOD RIVER DISTILLERS, Inc.',
'CRUSH WINES', 'POVERTY LANE ORCHARDS', 'DJINN SPIRITS LLC',
'MOONLIGHT MEADERY', 'TALL SHIP DISTILLERY LLC',
'FORTUNE WINE BROKERS LLC', 'BLACK COVE BEVERAGES', 'VINEXTRA INC',
'SURVILLE ENTERPRISES CORP', 'JEWELL TOWNE VINEYARDS',
'SWEETWATER FARM', 'ALTAMAR BRANDS LLC', 'CANDIA VINEYARDS',
'INCREDIBREW INC', 'ALISA CARR BEVERAGES',
'STELLAR IMPORTING CO LLC', 'FULCHINO VINEYARD INC',
'IRA GOLDMAN AND WILLIAMS, LLP', 'Circa Wines',
'VINEDREA WINES LLC', 'BLACK PRINCE DISTILLERY INC',
'VINEYARD BRANDS LLC', 'THE IMPORTED GRAPE LLC',
'WALPOLE MTN VIEW WINERY', 'GILMANTON WINERY & VINEYARD',
'HAUNTING WHISPER VYDS', 'STAR INDUSTRIES INC.',
'LOYAL DOG WINERY', 'R.P.IMPORTS INC', 'THE PIERPONT GROUP LLC',
'APPOLO VINEYARDS LLC', 'BLACK ROCK SPIRITS LLC',
'CENTEUR IMPORTS LLC', 'HIGHLAND WINE MERCHANTS LLC',
'AMERICAN SPIRITS EXCHANGE', 'UNCORKED', 'BRONCO WINE COMPANY',
'MILTONS DISTRIBUTING CO', 'TRUETT HURST', 'LAUREATE IMPORTS CO',
'FANTASY FINE WINES CORP', 'AAPER ALCOHOL & CHEMICAL CO',
'SILVER MOUNTAIN CIDERS', 'CAPSTONE INTERNATIONAL',
'FLAVOR ESSENCE INC'], dtype=object)

vendor_sales_summary['Description'].unique()

array(['Jack Daniels No 7 Black', "Tito's Handmade Vodka",
      'Absolut 80 Proof', ..., 'Crown Royal Apple',
      'Concannon Glen Ellen Wh Zin', 'The Club Strawbry Margarita'],
      shape=(9651,), dtype=object)

vendor_sales_summary['Volume'] =
    vendor_sales_summary['Volume'].astype('float')

vendor_sales_summary.fillna(0, inplace = True)

vendor_sales_summary['VendorName'] =
    vendor_sales_summary['VendorName'].str.strip()

vendor_sales_summary.dtypes

VendorNumber          int64
VendorName            object
Brand                 int64
Description           object
PurchasePrice         float64
ActualPrice           float64
Volume               float64
TotalPurchaseQuantity int64
TotalPurchaseDollars  float64
TotalSalesQuantity    float64
TotalSalesDollars     float64
TotalSalesPrice       float64
TotalExciseTax        float64
FreightCost           float64
dtype: object

vendor_sales_summary['GrossProfit'] =
    vendor_sales_summary['TotalSalesDollars'] -
    vendor_sales_summary['TotalPurchaseDollars']

```


vendor_sales_summary

	VendorNumber	VendorName	Brand	Description	PurchasePri
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89
...	...	...	...	...	...
10687	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71

10692 rows × 15 columns

```
vendor_sales_summary['GrossProfit'].min()
```

-52002.780000000006

```
vendor_sales_summary['ProfitMargin'] = (vendor_sales_summary['GrossProfit'] /  
    vendor_sales_summary['TotalSalesDollars'])
```

vendor_sales_summary

	VendorNumber	VendorName	Brand	Description	PurchasePri
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24

	VendorNumber	VendorName	Brand	Description	PurchasePri
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89
...	...	...	...	...	...
10687	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71

10692 rows × 16 columns

```

vendor_sales_summary['StockTurnover'] =
    vendor_sales_summary['TotalSalesQuantity']/vendor_sales_summary['TotalPurchaseQuantity']

vendor_sales_summary['SalestoPurchaseRatio'] =
    vendor_sales_summary['TotalSalesDollars']/vendor_sales_summary['TotalPurchaseDollars']

cursor = conn.cursor()

cursor.execute("""CREATE TABLE vendor_sales_summary (
    VendorNumber INT,
    VendorName VARCHAR(100),
    Brand INT,
    Description VARCHAR(100),
    PurchasePrice DECIMAL(10,2),
    ActualPrice DECIMAL(10,2),
    Volume INT,
    TotalPurchaseQuantity INT,
    TotalPurchaseDollars DECIMAL(15,2),
    TotalSalesQuantity INT,
    TotalSalesDollars DECIMAL(15,2),
    TotalSalesPrice DECIMAL(15,2),
    TotalExciseTax DECIMAL(15,2),
    FreightCost DECIMAL(15,2),
    GrossProfit DECIMAL(15,2),
    ProfitMargin DECIMAL(15,2),
    StockTurnover DECIMAL(15,2),
    SalesToPurchaseRatio DECIMAL(15,2),
    PRIMARY KEY (VendorNumber, Brand)
);
""")

```

```
<sqlite3.Cursor at 0x213b8218a40>
```

```
pd.read_sql('select * from vendor_sales_summary', conn)
```

	VendorNumber	VendorName	Brand	Description	PurchasePrice	Actual
--	--------------	------------	-------	-------------	---------------	--------

```
vendor_sales_summary.to_sql('vendor_sales_summary', conn, if_exists =  
    'replace', index = False)
```

10692

vendor_sales_summary

	VendorNumber	VendorName	Brand	Description	PurchasePri
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89
...	...	...	...	...	...
10687	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32
10688	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39
10689	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74
10690	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47
10691	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71

10692 rows × 18 columns

```
import sqlite3  
import pandas as pd  
import logging  
from ingestion_db import ingest_db  
  
logging.basicConfig(  
    filename = 'logs/get_vendor_summary.log',  
    level = logging.DEBUG,  
    format = '%(asctime)s - %(levelname)s - %(message)s',
```

```

        filemode = 'a'
    )

def create_vendor_summary(conn):
    '''This function will merge the different tables to get the overall
        vendor summary and adding new columns in the resultant data'''
    vendor_sales_summary = pd.read_sql_query("""WITH FreightSummary AS (
        SELECT
            VendorNumber,
            SUM(Freight) AS FreightCost
        FROM vendor_invoice
        GROUP BY VendorNumber
    ),

    PurchaseSummary AS (
        SELECT
            p.VendorNumber,
            p.VendorName,
            p.Brand,
            p.Description,
            p.PurchasePrice,
            pp.Price AS ActualPrice,
            pp.Volume,
            SUM(p.Quantity) AS TotalPurchaseQuantity,
            SUM(p.Dollars) AS TotalPurchaseDollars
        FROM purchases p
        JOIN purchase_prices pp
            ON p.Brand = pp.Brand
        WHERE p.PurchasePrice > 0
        GROUP BY p.VendorNumber, p.VendorName, p.Brand, p.Description,
            p.PurchasePrice, pp.Price, pp.Volume
    ),

    SalesSummary AS (
        SELECT
            VendorNo,
            Brand,
            SUM(SalesQuantity) AS TotalSalesQuantity,
            SUM(SalesDollars) AS TotalSalesDollars,
            SUM(SalesPrice) AS TotalSalesPrice,
            SUM(ExciseTax) AS TotalExciseTax
        FROM sales
        GROUP BY VendorNo, Brand
    )

    SELECT
        ps.VendorNumber,
        ps.VendorName,
        ps.Brand,
        ps.Description,
        ps.PurchasePrice,
        ps.ActualPrice,
        ps.Volume,
        ps.TotalPurchaseQuantity,
        ps.TotalPurchaseDollars,
        ss.TotalSalesQuantity,

```

```

        ss.TotalSalesDollars,
        ss.TotalSalesPrice,
        ss.TotalExciseTax,
        fs.FreightCost
FROM PurchaseSummary ps
LEFT JOIN SalesSummary ss
    ON ps.VendorNumber = ss.VendorNo
    AND ps.Brand = ss.Brand
LEFT JOIN FreightSummary fs
    ON ps.VendorNumber = fs.VendorNumber
ORDER BY ps.TotalPurchaseDollars DESC""",conn)

return vendor_sales_summary

def clean_data(df):
    '''this function will clean the data'''
    # changing datatype to float
    df['Volume'] = df['Volume'].astype('float')

    # filling missing value with 0
    df.fillna(0, inplace = True)

    #removing spaces from categorical columns
    df['VendorName'] = df['VendorName'].str.strip()
    df['Description'] = df['Description'].str.strip()

    #creating new columns for better analysis
    vendor_sales_summary['GrossProfit'] =
        vendor_sales_summary['TotalSalesDollars'] -
        vendor_sales_summary['TotalPurchaseDollars']
    vendor_sales_summary['ProfitMargin'] =
        (vendor_sales_summary['GrossProfit'] /
        vendor_sales_summary['TotalSalesDollars'])
    vendor_sales_summary['StockTurnover'] =
        vendor_sales_summary['TotalSalesQuantity']/vendor_sales_summary['TotalPurchaseQuantity']
    vendor_sales_summary['SalestoPurchaseRatio'] =
        vendor_sales_summary['TotalSalesDollars']/vendor_sales_summary['TotalPurchaseDollars']

    return df

if __name__ == '__main__':
    # creating database connection
    conn = sqlite3.connect('inventory.db')

    logging.info('Creating Vendor Summary Table.....')
    summary_df = create_vendor_summary(conn)
    logging.info(summary_df.head())

    logging.info('Cleaning Data.....')
    clean_df = clean_data(summary_df)
    logging.info(clean_df.head())

    logging.info('Ingesting data.....')
    ingest_db(clean_df, 'vendor_sales_summary', conn)
    logging.info('Completed')

```

